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Lies, Damned Lies, and Evals

How to trust your agents when the stakes are high

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Lower cost. The retrieval failure remained.

OBSERVED COST

−28%

on the affected task family

PRIMARY EVALUATION SCORE

Δ +0.0048 t = 0.27

ELAPSED TIME

+3.9%

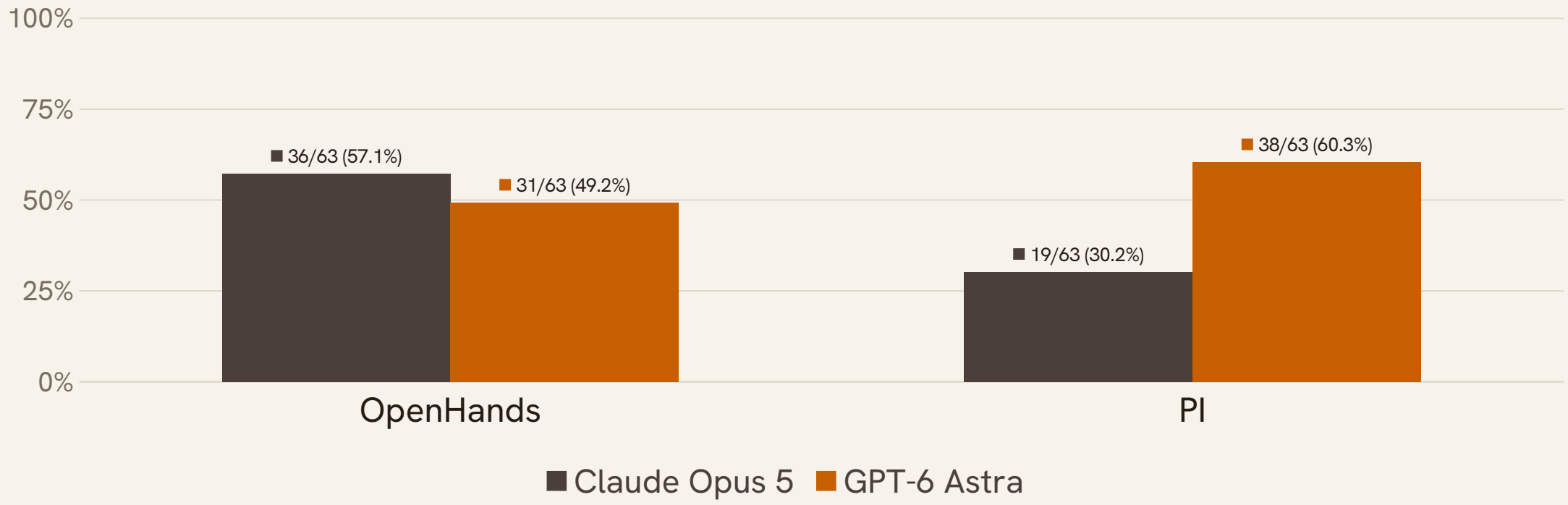
Search missed relevant evidence → Agent trusted the empty result

The retrieval failure remained. Little evidence of score improvement does not establish equivalent quality.

Jarmak · Engineering Reliable Coding Agents, ch. 18

The ranking changes with the harness

Same 63 tasks requiring no GPU. Different harness.



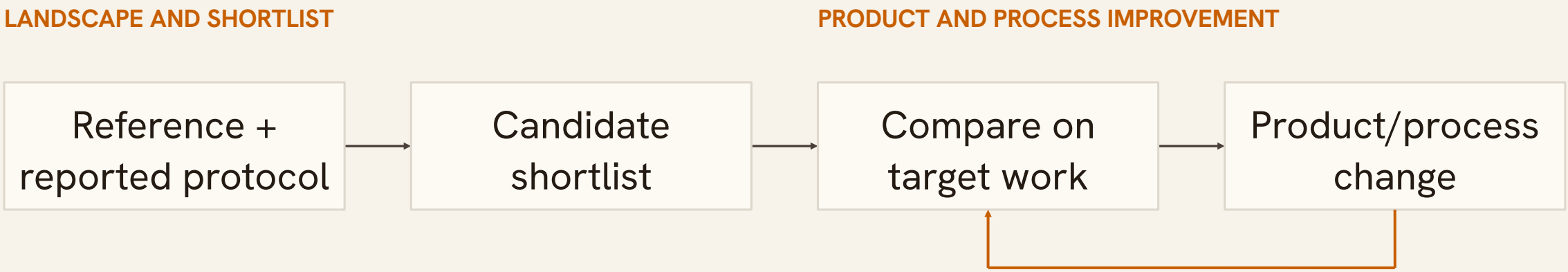
Compare the configuration you will actually run.

One scored run per task / configuration, after infrastructure retries.

Li et al. · Finding the Right Fit Hardy et al. · Agent Evaluation Reliability

Public benchmarks and internal evaluation

Benchmark: a standardized reference for comparison



Results depend on the task mix and evaluation protocol.

Diagnose, change and retest.
Communicate value against a baseline.

Internal benchmarks exist. Public benchmarks can be used internally.

Public and internal describe visibility, not exclusive purposes. The decision determines what evidence we need.

Jarmak · Engineering Reliable Coding Agents, chs. 2–3

Design the experiment around the decision

Report quality, reliability, people's time and total cost together.

TASKS

Workload coverage

Family A	task i
Family B	task j
Family C	task k

REPEATS

Variation between runs

Same task	run 1
Same task	run 2
Same task	run 3

CONFIGURATIONS

How results depend on the pairing

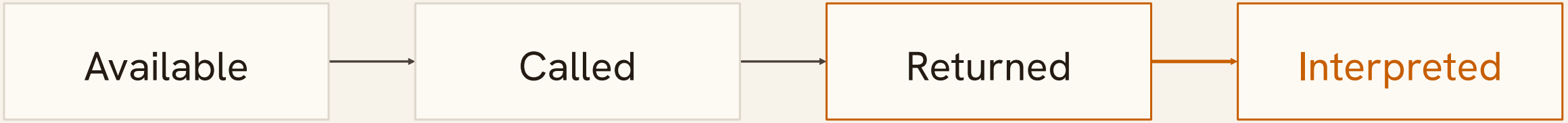
	Harness A	Harness B
Model A	A × A	A × B
Model B	B × A	B × B

Total operating cost ÷ Independently accepted outcomes

Repeated runs are not independent tasks. A better relative score does not establish that either system is reliable enough.

Jarmak · Engineering Reliable Coding Agents, chs. 1–2 Hardy et al. · Agent Evaluation Reliability

Trace where evidence becomes a conclusion



RECORD THE SEARCH

Where and how you searched

Gaps or incomplete results

What you couldn't check

SEARCH-LIMITS TEST

Same empty result, make the search limits explicit

EVIDENCE TEST

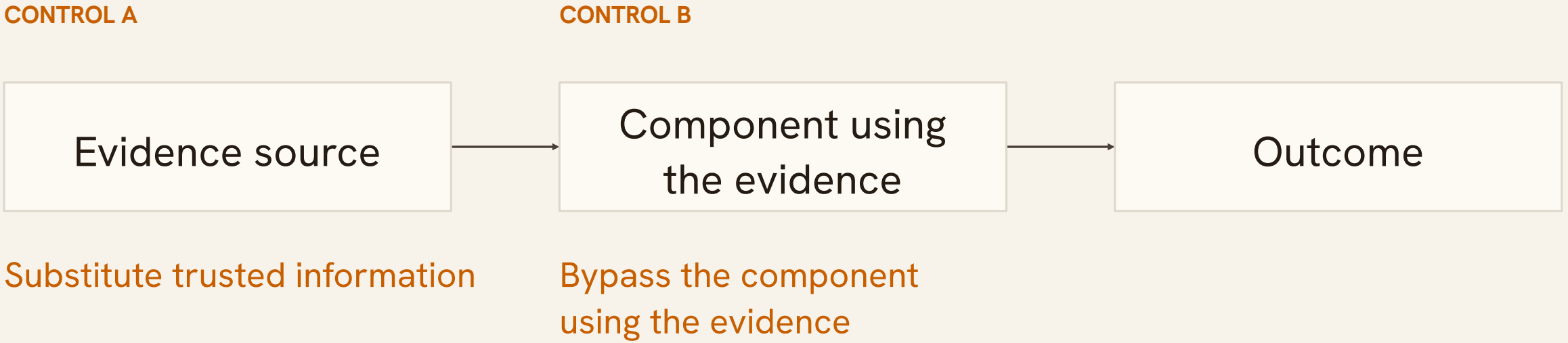
Same agent and settings, supply known relevant evidence

A trace tells us where to investigate. These proposed tests have not yet demonstrated a repair.

Jarmak · Engineering Reliable Coding Agents, chs. 2 & 18

Better evidence can still go unused

Test the evidence and the agent’s use of it separately.



Deployment comparison

Keep non-use and failed attempts in the result.

Simulation result; production effects remain untested. Tool use alone does not identify the tool’s causal effect.

Alzahrani · Verify Claims, Not Scores Jarmak · Engineering Reliable Coding Agents, ch. 2

Audit the scoring rule and the answer key

RECORDED OUTPUT

295

files returned



6 reference files found

289 outside the reference set

SCORING RULE

RECALL

1.0

PRECISION

0.020339

about 2.0%

ACTUAL F-BETA SCORE

0.025295

beta = 0.5

PROPOSED CHALLENGES

Known-good answer

Empty answer

Return-everything answer

Valid alternative absent from the key

AUDITABLE RESULT RECORD

Scoring rule
Reference version
Run identity

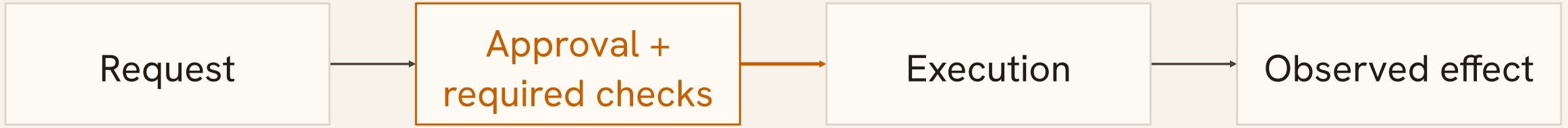
One output, measured against one reference set. The score was precision-sensitive; intent and prevalence are not established.

Jarmak · CodeProbe trial e5d7a4e7 / commit 2495927

Test whether the gate changes execution

My approval-queue audit found paths that allowed execution when required components were missing.

PROPOSED ENFORCEMENT TEST



DELIBERATELY BREAK A CONTROL

- Reject or cancel
- Remove a required component
- Change the proposal after approval

MEASURE OUTCOMES SEPARATELY

- Authorized completion
- Safe stop
- People’s repair work

Deliberately break a control next. Also measure false blocks on legitimate work; this audit did not establish unauthorized production writes.

Jarmak · Engineering Reliable Coding Agents, chs. 4 & 17

Choose what runs automatically, then measure it

Retrieve existing rules; clarify unresolved requirements; encode where possible.

DEFECT DETECTION

What consequential errors get through?

REVIEW LOAD

What good work goes to review?

SHARE HANDLED AUTOMATICALLY

What fraction needs no human review?

ONE OPERATING CONSEQUENCE

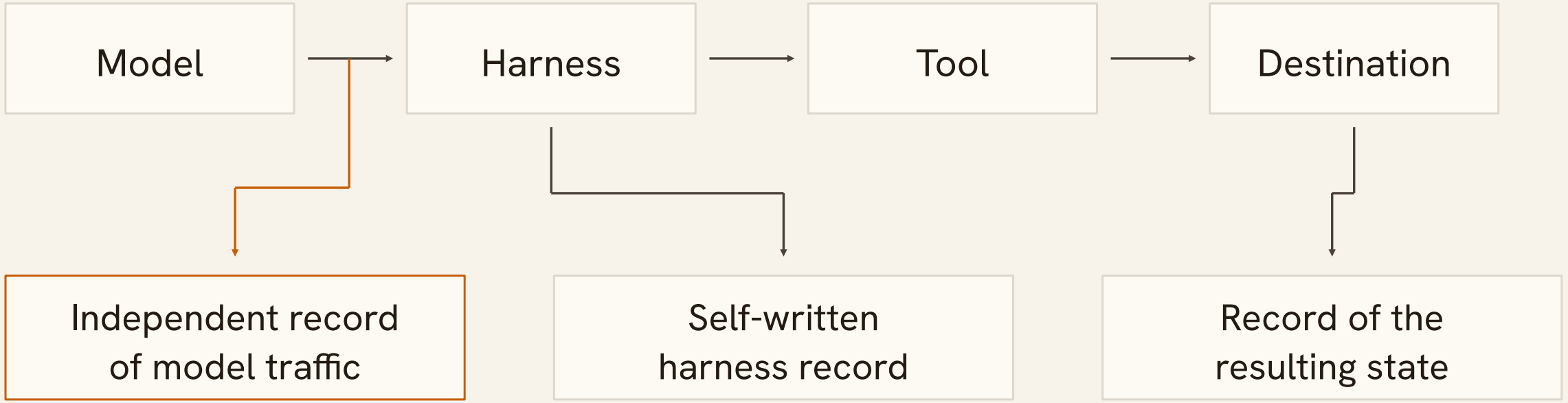
1%	×	1,000,000	=	10,000
false-alarm rate		legitimate actions		reviews

Worked example, not measured traffic. Samples enriched for suspected errors do not estimate production error rates.

Liu et al. · Auditing and Repairing LLM-as-Judge Failures Sehanobish et al. · From Benchmarks to Production (FLINT)

Some evidence needs another observer

For security claims, test missed detections, false alarms and blind spots.



Own record passed the hash check

28/28

Independent record comparison detected

28/28

28 deliberately introduced changes. Detection beyond this scope is untested; model traffic does not prove execution or destination state.

A retrieval upgrade: the experiment I would run

PROPOSED EVALUATION WORKFLOW

Matched target-work comparison

↓

Independent challenge

↓

Run alongside current work.
Test state changes in isolation.

Can an independent challenger find the decisive evidence?

AFTER DEPLOYMENT

Completion, repair, latency, cost
Audit accepted work and failures
Check traces against real outcomes
Watch workload and version changes
Owner, intervention, regression tests

Do scores predict real outcomes?

Set restriction or rollback triggers before deployment. The monitoring loop and independent-review tests are proposed.

Jarmak, chs. 16–17 · Google SRE: monitoring and alerting

Make the next autonomy decision explicit

PROPOSED AUTONOMY DECISION

Additional action + operating scope + human check removed

EVIDENCE BAR

Representative repeated outcomes
Tolerable failures and total burden

VALIDATED EVALUATOR + EXECUTION CONTROLS

Accept good work, detect defects
and enforce the operating boundary

Human owner: acceptable risk and conflicting business values

Expand within the tested boundary	Retain review	Stop and escalate

Stop for missing evidence, a failed gate or changed scope.

An action-specific standard to test. Procedures support the work; they do not establish reliability or supply missing evidence.

Evaluation procedures and references

Coding Agent Workflows



github.com/sjarmak/coding-agent-workflows

THREE PROCEDURES

Evaluation design

Harness traceability

Reliability procedures

BOOK AND PAPER

Engineering Reliable Coding Agents

Evaluating and Operating the System Around the Model

sjarmak.ai/books/engineering-reliable-coding-agents

arxiv.org/abs/2608.13867